Interpretation of Verification Results for Existential and Hybrid Properties

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For existential properties, the interpretation is dual to the interpretation used for universal properties. In CTL, an existential requirement can be written, for example as $EF(\gamma)$, which requires that there exists at least one path along which a state satisfying γ is eventually reached. If NuSMV reports that the learned Moore machine satisfies such a requirement, MELA checks whether the existential witness is supported by the PTA. To do so, MELA verifies the same CTL property on the PTA encoded in NuSMV. If the PTA also satisfies the property, then the satisfying behaviour is supported by the observed traces, and MELA reports the requirement as satisfied. If the PTA violates the property, then the witness in the learned Moore machine is attributed to state merging, and MELA returns a violation.

The existence of an execution satisfying an LTL formula ψ, is checked through the universal complement. Specifically, MELA verifies $\neg\psi$ as an LTL specification. If NuSMV violates $\neg\psi$, the returned counterexample is a witness execution satisfying ψ, and the existential requirement is satisfied on the learned Moore machine. MELA then performs the same check on the PTA. If $\neg\psi$ is also violated on the PTA, the existential witness is supported by the observed traces, and MELA reports the requirement as satisfied. If $\neg\psi$ is satisfied on the PTA, then the existential witness exists only in the learned Moore machine and is attributed to state merging.

If an existential property is violated on the learned Moore machine, then no execution in the learned machine satisfies the required behaviour. Since the learned machine contains the behaviours represented by the PTA, the absence of such an execution in the learned machine means that the PTA cannot provide an observed witness either. In this case, MELA reports the requirement as violated. For existential implication properties, MELA applies the same vacuity principle as for universal implication properties: satisfaction is considered vacuous when the condition that should trigger the required behaviour is never reached in the relevant witness structure.

Hybrid properties combine universal and existential reasoning. In CTL, a typical example is $AG(\alpha \rightarrow EF(\beta))$, which states that whenever an α-state is reached, there must exist a continuation from that state that eventually reaches a β-state. For such properties, MELA applies the universal interpretation to the universally quantified part and the existential interpretation to the existentially quantified part. Thus, MELA first checks whether the antecedent α is reachable to determine whether satisfaction is vacuous. If the property is violated, MELA checks whether the PTA also violates the property, since the violation may have been introduced by state merging. If the property is satisfied, MELA checks whether the existential witnesses required by the property are also present in the PTA. If these witnesses are present, MELA reports non-vacuous satisfaction; otherwise, MELA reports a violation.

For LTL hybrid properties, MELA uses the same principle, but existential reasoning is expressed through complement checking. Universal LTL properties are interpreted directly as NuSMV satisfaction or violation results. Existential LTL properties are interpreted through violations of their complements, where the counterexample returned by NuSMV serves as the witness execution. MELA then checks the corresponding property on the PTA to determine whether the reported violation or witness is supported by the observed traces or introduced by state merging.